Interacting Multipoles of Rank 2 and 3:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta,\gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha,\gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha,\beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta,\gamma) \cdot \delta(\delta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta,\delta) \cdot \delta(\gamma,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta,\epsilon) \cdot \delta(\gamma,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha,\gamma) \cdot \delta(\delta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha,\delta) \cdot \delta(\gamma,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha,\epsilon) \cdot \delta(\gamma,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha,\beta) \cdot \delta(\delta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha,\delta) \cdot \delta(\beta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha,\epsilon) \cdot \delta(\beta,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha,\beta) \cdot \delta(\gamma,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha,\gamma) \cdot \delta(\beta,\epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha,\epsilon) \cdot \delta(\beta,\gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha,\beta) \cdot \delta(\gamma,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha,\gamma) \cdot \delta(\beta,\delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha,\delta) \cdot \delta(\beta,\gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \end{bmatrix} \tag{1}$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta\left(\delta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\delta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\delta,\delta\right) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \end{bmatrix} \tag{2a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \end{bmatrix} \tag{2b}$$

$$\xRightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\delta} \\ -30 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\delta} \\ -30 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\delta} \\ -30 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\delta} \\ -30 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\epsilon} \\ -30 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \end{bmatrix} \tag{2c}$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} -945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{zzz} + 105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{z\delta\delta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{zzz} \\ -15 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{\beta\delta\delta} - 30 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\delta\delta} - 30 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\delta\delta} - 30 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\gamma\gamma} - 30 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\gamma\gamma} - 30 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} +105 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} + 105 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ +105 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} + 105 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ +105 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} + 105 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\delta\delta} - 30 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\delta\delta} - 30 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ -30 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} - 30 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ -30 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \end{bmatrix} \tag{3b}$$

$$\xRightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} +315 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} + 315 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\delta\delta} - 30 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\delta\delta} - 30 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ -90 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \end{bmatrix} \tag{3c}$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} +315 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{zzz} + 315 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{zzz} \\ -15 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\delta\delta} - 30 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\delta\delta} - 30 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ -90 \cdot R_z^5 \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\alpha\alpha} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot [+0] \tag{4b}$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = +0 \tag{5}$$